Suman Sourav

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Summary

My research explores the intersection of theory and systems application, focusing distributed dynamic systems, and communication networks such as satellite networks, IoT and edge-cloud networks, cyber-physical systems, etc. During my PhD, I received the *Research Achievement Award* from the School of Computing, NUS.

Scientific Focus Areas

Distributed Systems; Satellite Networks; IoT and Edge-Cloud Systems; Network Communication Algorithms; CPS Security.

Work Experience

Aalborg University — Copenhagen, Denmark
Assistant Professor — *March 2025 - Present*

Singapore University of Technology and Design — Singapore
Research Fellow II, Mentor: Dr. Binbin Chen — *March 2022 - March 2025*

Illinois Advanced Research Center, University of Illinois at Urbana-Champaign (UIUC) — Singapore
Adjunct Postdoctoral Researcher — *March 2022 - January 2025*

Illinois Advanced Research Center, University of Illinois at Urbana-Champaign (UIUC) — Singapore
Postdoctoral Researcher, Mentor: Dr. Binbin Chen — *January 2020 - March 2022*

National University of Singapore — Singapore
Research Assistant, Advisor: Dr. Seth Gilbert — *July 2018 - December 2019*

Indian Institute of Technology — Chennai, India
Visiting Researcher, Host: Dr. John Augustine — *July 2019 - August 2019*

Education

National University of Singapore — Singapore
Doctor of Philosophy in Computer Science; GPA: 4.4/5.0 — *January 2014 - November 2019*
Thesis: Latency, Conductance and the Role of Connectivity in Graph Problems
Advisor: Dr. Seth Gilbert

Indian Statistical Institute — Kolkata, India
Master of Technology (M.Tech) in Computer Science; Percentage: 82.63/100 — *June 2011 - July 2013*
Thesis: A Quantitative Analysis of the Bitcoin Transaction Graph
Advisor: Dr. Palash Sarkar

Veer Surendra Sai University of Technology — Burla, India
Bachelor of Technology (B.Tech.) in Computer Science and Engineering; GPA: 8.71/10.0 — *May 2007 - May 2011*
Thesis: Better Communication through CrossCube Ring Connected Networks
- **University Topper** in 5th, 7th and 8th Semester

Advisor: Dr. C.R. Tripathy

Research Grants

- EstateSense: A Scalable Distributed Sensing System for Smart Estates", funded by MND Cities of Tomorrow. **Grant Amount:** SGD 2.89 Million). **Role:** Work Package Leader.[1]
- Enhancing Power System Resilience through Distributed Intelligence and Adaptive Infrastructure, funded by Singapore Energy Market Authority (EMA). **Grant Amount:** SGD 2.2 million. **Role:** Work Package Co-Lead. [1]
- Towards Practical Attestation Solutions for Countering Advanced Attacks to Industrial Control Systems, funded by National Satellite of Excellence, Design Science and Technology for Secure Critical Infrastructure (NSoE DeST-SCI). **Grant Amount:** SGD 939000. **Role:** Collaborator.[1]
- Leveraging correlated unpredictability for enhancing system scalability and security, applied as SUTD Start-up Research Grant. **Grant Amount:** SGD 50000. **Role:** Proposal Writer.[1]

[1]Postdoctoral Researchers cannot be PI/Co-PI in Singapore.

Academic Awards and Achievements

- **Research Achievement Award** for outstanding research performance at NUS, School of Computing for Academic Year 2017-2018.
- Was one of the **100 qualified young researchers** from all over the world to be selected for the **7th Heidelberg Laureate Forum** 2019.
- Was one of the **20 alumni** to be invited back to the Heidelberg Laureate Forum from all previous attendees on the 10th anniversary of the Heidelberg Laureate Forum in 2023.
- Was selected to attend and participate in the **Global Young Scientists Summit** 2024, held annually in Singapore and conducted by the National Research Foundation, Singapore.
- **Represented India** in a five-member delegation to Egypt in the 39th International Youth Camp held in Alexandria, Egypt (Aug - Sept, 2007).
- Was nominated to **represent India** in a 100-member youth delegation to China by the Ministry of Youth Affairs and Sports (June 2007).

Select Publications in refereed Journals in reverse chronological order

- Wang P.; Chen B.; **Sourav S**.; Li H., "One-Pass & Smart-Search: A Solver for Minimizing Data Delivery Time over Time-Varying Networks", IEEE Transactions on Mobile Computing (IEEE TMC) Journal, (Accepted and to appear), doi: 10.1109/TMC.2026.3715422, Journal Core Ranking: A^*.
- Wang P.; **Sourav S**.; Chen B.; Li H., Wang F.; Zhang F. "The Throughput Gain of Hypercycle-level Resource Reservation for Time-Triggered Communication", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems Journal, (Accepted and to appear), Journal Core Ranking: A^* equivalent.
- Wang P.; **Sourav S**.; Chen B.; Li H., "An SFC-constrained Max-Flow Solver for Satellite Networks using Flexible Function-Time Expanded Graph", IEEE Transactions on Mobile Computing (IEEE TMC) Journal, vol. 25, no. 3, pp. 4103-4120, Mar. 2026, doi: 10.1109/TMC.2025.3623456, Journal Core Ranking: A^*.
- Wang P.; **Sourav S**.; Chen B.; Li H., "FlexSatIoE: Flexible Routing and Buffering for Satellite Networks Enabled Internet of Everything Applications": IEEE Internet of Things Journal (IEEE IOTJ), vol. 13, no. 1, pp. 196-211, 1 Jan.1, 2026, doi: 10.1109/JIOT.2025.3594890, Journal Core Ranking: A^* equivalent.
- Zhou H.; **Sourav, S.**; Chen B., Yu K., "SRLR: Symbolic Regression based Logic Recovery to Counter Programmable Logic Controller Attacks", IEEE Transactions on Information Forensics and Security (TIFS) Journal, , vol. 20, pp. 12491-12506, 2025, doi: 10.1109/TIFS.2025.3634027, Journal Core Ranking: A^*.
- **Sourav, S.**; Chen, B. ,"Exposing Hidden Attackers in Industrial Control Systems using Micro-distortions": IEEE Transactions on Smart Grid, vol. 15, no. 2, pp. 2089-2101, March 2024, doi: 10.1109/TSG.2023.3300710, Journal Core Ranking: A^* equivalent.
- Gilbert, S.; Robinson, P.; **Sourav, S.**[2], "Leader Election in Well-Connected Graphs": Algorithmica, pp. 1029–1066, Vol. 85, Apr 2023, doi: 10.1007/s00453-022-01068-x, Journal Core Ranking: A^*.
- Augustine, J.; Gilbert, S.; Kuhn, F.; Robinson, P.; **Sourav, S.**[2], "Latency, Capacity and Distributed MST": Journal of Computer and System Sciences (JCSS), pp. 1–20, Vol. 126, Jul 2022 doi: 10.1016/j.jcss.2021.11.006, Journal Core Ranking: A^*.
- Augustine, J.; Choudhary, K.; Cohen, A.; Peleg, D.; Sivasubramaniam, S.; **Sourav, S.**[2], "Distributed Graph Realizations": IEEE Transactions on Parallel and Distributed Systems (TPDS), Vol. 33, no. 6, pp. 1321–1337, 1 June 2022, doi: 10.1109/TPDS.2021.3104239, Journal Core Ranking: A^*.
- **Sourav, S.**; Robinson, P.; Gilbert, S., "Slow Links, Fast Links, and the Cost of Gossip": IEEE Transactions on Parallel and Distributed Systems (TPDS), Vol. 30, no. 9, pp. 2130–2147, 1 Sept. 2019, doi: 10.1109/TPDS.2019.2905568, Journal Core Ranking: A^*.

Select Publications in refereed Conferences in reverse chronological order

- Debadarshini J.*; **Sourav S.***; Singh L.; Kumar S.S.; "On Overcoming Compute and Memory Constraints for Distributed Machine Learning at the IoT Edge", IEEE Conference on Local Computer Networks (LCN 2026), (Accepted and to appear), Conference Core Ranking: *B*.
  * ***Both Authors Contributed Equally. The paper has also been nominated as a candidate for the best paper award.***
- Debadarshini J.; **Sourav S**.; Gurusamy M., "ReClub: Real-Time Cluster-Formation for UAV-Assisted Scalable Data-Collection", 2026 IEEE Wireless Communications and Networking Conference (WCNC 2026), pp. 1-6, doi: 10.1109/WCNC65185.2026.11555742., Conference Core Ranking: *B*.
- Choudhary V.; Wang P.; **Sourav, S.**; Chen, B. ,"Enhancing Distributed Data Processing Throughput in IoT Systems using Optimized Task Placement": IEEE International Conference on Distributed Computing Systems (ICDCS 2024), pp. 1474-1475, doi: 10.1109/ICDCS60910.2024.00158, Conference Core Ranking: *A*.
- Wang P.; **Sourav, S.**; Li H.; Chen, B. ,"One Pass is Sufficient: A Solver for Minimizing Data Delivery Time over Time-varying Networks": IEEE International Conference on Computer Communications (INFOCOM 2023), pp. 1-10, doi: 10.1109/INFOCOM53939.2023.10228959, , Conference Core Ranking: A^*.
- **Sourav, S.**; Biswas,P.; Mohanraj,V., Chen, B. ; Mashima, D.;"Machine Learning Assisted Bad Data Detection for High-throughput Substation Communication": IEEE International Conference on Communications (ICC 2023), pp. 4131-4137, doi: 10.1109/ICC45041. 2023.10278910, Conference Core Ranking: *B*.

[2] In the corresponding papers, authors names here are alphabetically ordered as is common practice in Distributed Computing research.

- **Sourav, S.**; Biswas,P.; Chen, B. ; Mashima,D.,"Detecting Hidden Attackers in Photovoltaic Systems Using Machine Learning": IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (IEEE SmartGridComm 2022), pp. 360–366, doi: 10.1109/SmartGridComm52983.2022.9960965, Conference Core Ranking: *B* equivalent.
- **Sourav, S.**; Chen, B.,"Distort to Detect, not Affect: Detecting Stealthy Sensor Attacks with Micro-distortion": IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (IEEE SmartGridComm 2021), pp. 412–418, doi: 10.1109/SmartGridComm51999.2021.9632311, Conference Core Ranking: *B* equivalent.
- Augustine, J.; Gilbert, S.; Kuhn, F.; Robinson, P.; **Sourav, S.**[2], "Latency, Capacity and Distributed MST": IEEE International Conference on Distributed Computing Systems (ICDCS, 2020), pp. 157–167, doi: 10.1109/ICDCS47774.2020.00055, Conference Core Ranking: *A*.
- Augustine, J.; Choudhary, K.; Cohen, A.; Peleg, D.; Sivasubramaniam, S.; **Sourav, S.**[2], "Distributed Graph Realizations": IEEE International Parallel and Distributed Processing Symposium (IPDPS, 2020), pp. 158–167, doi: 10.1109/IPDPS47924.2020.00026 , Conference Core Ranking: *A*.
- Gilbert, S.; Robinson, P.; **Sourav, S.**[2], "Leader Election in Well-Connected Graphs": ACM Symposium on Principles of Distributed Computing (PODC, 2018), pp.227–236, doi: 10.1145/ 3212734.3212754, Conference Core Ranking: A^*.
- **Sourav, S.**; Robinson, P.; Gilbert, S., "Slow Links, Fast Links, and the Cost of Gossip": IEEE International Conference on Distributed Computing Systems (ICDCS, 2018), pp. 786–796, doi: 10.1109/ICDCS.2018.00081, Conference Core Ranking: *A*.
  Nominated as a candidate for the best paper award.
- Gilbert, S.; Robinson, P.; **Sourav, S.**[2], "Brief Announcement: Gossiping with Latencies": ACM Symposium on Principles of Distributed Computing (PODC, 2017), pp. 255–257, doi: 10.1145/3087801.3087846, Conference Core Ranking: A^*.

Book Chapters

- **Sourav, S.**; Debadarshini J.; Jindal A.; Chen B.; "Detecting Stealthy Attackers Hiding in Edge-IoT Systems by Using Micro-distortions", Edge Intelligence for IoT and Healthcare: Advances and Applications, book to be published by Springer as part of the Transactions on Computer Systems and Networks series. (Accepted)

Teaching Experience

- Distributed Systems as Asst. Professor, Aalborg University, *Fall 2026*.
- Coordinator for Master's Specialization course, *Fall 2026*.
- Models and Tools for Cyber-Physical Systems as Asst. Professor, Aalborg University, *Spring 2025 and Spring 2026*.
- Computer Networks as Guest Lecturer, SUTD, Singapore, *Jan 2023 - June 2023*.
- CS3230 Design and Analysis of Algorithms as Teaching Assistant at NUS SoC, *Jan 2017 - Jul 2017*.
- CS3230 Design and Analysis of Algorithms as Teaching Assistant at NUS SoC, *Jan 2016 - Jul 2016*.
- 'C and Data structures', and 'Operating Systems' as Assistant Professor at VSSUT, Burla, *Aug 2014 - Jan 2015*.

Student Supervision

At Aalborg University Copenhagen, Denmark
3 Master's Thesis students *Jan 2026 - Present*
4 Master's specialization students *Jun 2025 - Jan 2026*
6 Bachelor's Thesis students and 7 Bachelor's SW2 student's project *Jan 2026 - Present*

At Singapore University of Technology and Design Singapore
Co-supervised 3 PhD students and 2 Master's students *March 2022 - March 2025*

Illinois Advanced Research Center, University of Illinois at Urbana-Champaign (UIUC) Singapore
2 Research Engineers and 2 Research Assistants *January 2020 - March 2022*

Dissemination through Invited Talks, Panels, and Presentations

- (Upcoming) Workshop at the 13^{th} Heidelberg Laureate Forum, 2026 on 14.09.2026.
- **Keynote Talk** at International Workshop on Smart Computing for Smart Cities (SC2), IEEE WoWMoM Conference, Italy, on 16.06.2026.
- Invited Panel member at the International Workshop on Artificial Intelligence for Networked Drone and Sensor Applications (DroneSense-AI) 2026, IEEE WoWMoM Conference, Italy, on 16.06.2026.
- Invited Panel member at the AlumNode Panel Talk on Mentoring in Academia & Industry on 08.05.2025.
- University of New South Wales, Sydney, Australia on 23.02.2024 - Minimizing Data Delivery Time over Satellite Networks.
- SUTD Wireless Innovation Centre (SWIC), Singapore on 23.09.2023 - Minimizing Data Delivery Time over Satellite Networks.
- Invited for presentation at the Highlights of Algorithms (HALG 2022) - Latency, Capacity and Distributed MST.
- IIT Hydrabad, India, on 06.12.2022 - Leader Election in Well-Connected Graphs.
- Xidian University, China, on 01.12.2022 - Latency Conductance and the Role of Connectivity in Graph Problems.
- ISI Kolkata, India on 21.11.2022 - Latency, Capacity, and Distributed MSTs.
- Heidelberg Laureate Forum, Germany, on 26.09.2019 - Gossip with Latencies.
- SoC Research Talk at NUS, Singapore on 11.10.2017 - Bounds on Distributed Information Spreading in Networks with Latencies.

ACADEMIC SERVICE

- **Mentor** in HLF Alumnode (2021-Present) and **Mentor** in Danish Data Science Academy Mentorship Programme (2025-Present).
- **External Thesis Examiner**, 1 Master's student and 5 Bachelor's students at IT University, Copenhagen, and 1 Bachelor's student at the University of Southern Denmark, Odense.
- **Guest Editor** for the Special Issue titled "Advances in Routing and Scheduling Technology for Data Communications" in the journal Network (MDPI) (ISSN 2673-8732).
- **Program Committee member** for IEEE SmartGridComm-2026, IEEE SmartGridComm-2025, ICDCIT-2023.
- Session chair for Data Analysis and Computation in Smart Metering session at IEEE SmartGridComm-2022
- Reviewer for IEEE TMLCN-2026, IEEE TIFS-2026, IEEE IOTJ-2026, IEEE VTC-2026, IEEE TIFS-2024, IEEE TOC-2023, IEEE TSG-2023, IEEE TSG-2022, ICDCS-2022, IEEE TGCN-2021, ICDCS-2021, PODC-2021, DISC-2020, SIROCCO-2020, OPODIS-2019.

REFERENCES

- **Assoc. Prof. Binbin Chen**,
  Associate Head of Pillar (Innovation and Enterprise), SUTD, Singapore
  Deputy Director, Future Communications Research and Development Programme (FCP), Singapore
  Email - binbin_chen@sutd.edu.sg
- **Prof. Seth Gilbert**,
  Head of Department, Computer Science, NUS, Singapore.
  Email - seth.gilbert@comp.nus.edu.sg